a **Graded cross-modal similarity**

$\Delta_{\text{matched – same}} = 0.089$
$p = 7.7e-12$

$\Delta_{\text{same – other}} = 0.053$
$p = 2.6e-12$

Brain–CLIP cosine similarity

0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05

Matched stimulus
Same-cluster non-match
Different cluster

b **Geometry across all semantic clusters**

People & portraits (C7)
Mountain landscapes (C0)
Soups & hot beverages (C6)
Skiing & snow sports (C30)
Tennis action (C33)
Snakes (C14)
Dogs (C2)
Baseball (C21)
Sharks & marine life (C4)
Airplanes & aviation (C17)
Birds (C39)
Bridges & urban infrastructure (C24)
Consumer electronics (C36)
Stone bridges & viaducts (C8)
Libraries & auditoriums (C26)
Zebras (C27)
Trains & rail transport (C22)
Bathrooms & washrooms (C1)
Citrus & fruit (C34)
Shorebirds & wading birds (C19)
Water towers (C18)
Coastal landscapes (C9)
Wooden furniture (C5)
Ceramics & tableware (C28)
Rural fields (C35)
Buses & trucks (C16)
Classic vehicles (C31)
Textile accessories (C12)
Emergency vehicles (C23)
Sailboats & sailing (C15)
Flowers (C29)
Structural details & patterns (C20)
Religious architecture (C25)
Clocks & timers (C3)
Tools & hardware (C37)
Insects (C10)
Home appliances (C11)
Household containers & bins (C13)
Toiletries & personal care (C32)
Books, plaques & instruments (C38)

Matched − same cluster
Same cluster − different cluster
Hollow: FDR $q \geq 0.05$

0.000
0.025
0.050
0.075
0.100
0.125
0.150

Subject-level cosine-similarity margin

c **Robustness across datasets**

BMD ($n = 10$)
NSD ($n = 8$)
THINGS ($n = 3$)
GOD ($n = 5$)
BOLD5000 ($n = 4$)
deeprecon ($n = 3$)
NOD ($n = 30$)

Matched − same cluster
Same cluster − different cluster

0.00
0.05
0.10
0.15
0.20
0.25
0.30

Subject-level cosine-similarity margin